

Degeneracy of the Inverse Problem: Uncountable Solutions from Dynamics

1 The Forward Model

The flyvis graded-voltage ODE for each postsynaptic neuron i reads:

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij} \text{ReLU}(v_j(t)) + I_i(t) \quad (1)$$

where τ_i and V_i^{rest} are cell-type parameters, $\mathbf{W}_{ij}$ is the connectome-constrained synaptic weight from neuron j to i , and $I_i(t)$ is the external stimulus.

The inverse problem: given observed trajectories $\{v_i(t)\}_{t=0}^T$ and stimuli $\{I_i(t)\}$, recover the connectivity matrix $\mathbf{W}$, time constants τ , and resting potentials V^{rest} .

2 Where the Degeneracy is Hard-Coded

2.1 Step 1: Rearrange to a linear system per neuron

For each postsynaptic neuron i , define the presynaptic activity vector at time t :

$$h_j(t) = \text{ReLU}(v_j(t)) \quad (2)$$

and the right-hand side:

$$b_i(t) = \tau_i \frac{dv_i(t)}{dt} + v_i(t) - V_i^{\text{rest}} - I_i(t) \quad (3)$$

Let $d_i = |\mathcal{N}_i|$ denote the in-degree of neuron i . Then the ODE becomes a linear constraint on the incoming weights $\mathbf{w}_i = (\mathbf{W}_{i1}, \mathbf{W}_{i2}, \dots, \mathbf{W}_{id_i})$:

$$\sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij} h_j(t) = b_i(t) \quad \forall t \in \{1, \dots, T\} \quad (4)$$

Stacking over T timesteps yields a **linear system**:

$$\mathbf{H}_i \mathbf{w}_i = \mathbf{b}_i \quad (5)$$

where $\mathbf{H}_i \in \mathbb{R}^{T \times d_i}$ is the activity matrix (columns = presynaptic neuron activations over time).

2.2 Step 2: The null space

Any $\boldsymbol{\delta} \in \ker(\mathbf{H}_i)$ satisfies $\mathbf{H}_i \boldsymbol{\delta} = \mathbf{0}$. Therefore the perturbed weight vector $\mathbf{w}_i + \boldsymbol{\delta}$ produces **exactly the same dynamics** as $\mathbf{w}_i$:

$$\mathbf{H}_i(\mathbf{w}_i + \boldsymbol{\delta}) = \mathbf{H}_i \mathbf{w}_i + \underbrace{\mathbf{H}_i \boldsymbol{\delta}}_{=\mathbf{0}} = \mathbf{b}_i \quad (6)$$

The null space dimension is $\dim \ker(\mathbf{H}_i) = d_i - \text{rank}(\mathbf{H}_i)$.

Global bound from SVD. An SVD of the full population activity reveals that the entire circuit’s activity lives in a low-dimensional subspace. The computation subsamples to 982 neurons (every 14th of 13,741) and 8,000 timesteps (every 8th of 64,000), then computes the full SVD (all 982 singular values):

Metric	Value
Neurons (subsamped)	982 / 13,741
Timesteps (subsamped)	8,000 / 64,000
SVD	full (982 singular values)
Effective rank at 90% variance	1
Effective rank at 99% variance	45

Since each neuron i ’s presynaptic activity $\mathbf{H}_i$ is a submatrix of the global activity, $\text{rank}(\mathbf{H}_i) \leq 45$ regardless of the in-degree d_i . This gives a per-neuron bound:

$$\dim \ker(\mathbf{H}_i) \geq \max(0, d_i - 45) \quad (7)$$

and a global bound by summing over all 13,697 postsynaptic neurons:

$$\text{total null dim} = \sum_{i=1}^{N_{\text{post}}} \max(0, d_i - r) \quad (8)$$

where r is the effective activity rank. The in-degree distribution determines which neurons contribute:

In-degree range	Neurons	Null space contribution at rank 45
1–10	3,182	0 (fully identifiable)
11–20	3,917	0 (fully identifiable)
21–45	3,196	0 (fully identifiable)
46–100	2,843	bulk of null space
101–208	559	largest per-neuron null dimensions

Since the mean in-degree is only 32 and the activity rank is 45, most neurons have $d_i \leq 45$, making their null space contribution zero—their incoming weights are fully identifiable. The null space comes entirely from the minority of highly connected neurons (in-degree > 45). Why are some in-degrees so high? The 217 columns are not independent pipelines—they are laterally interconnected. A medulla neuron integrating signals from its spatial neighborhood receives edges from the same cell type in many surrounding columns (e.g., Mi1 in column 5 receives input from L1 in columns 4, 5, 6, and nearby neighbors). A neuron with in-degree 100 might receive input from 50 copies of one cell type spread across 50 columns.

Of the 13,741 total neurons, 44 are source-only (they send but never receive edges), leaving 13,697 postsynaptic neurons. These 44 source-only neurons belong to 4 cell types at the boundary of the 217-column retinotopic subset:

Cell type	Source-only	Neurons of this type	Notes
C3	1 / 217	centrifugal neuron	0.5% boundary
L1	9 / 217	lamina monopolar cell	4.1% boundary
R4	17 / 217	photoreceptor	7.8% boundary
Tm9	17 / 217	transmedullary neuron	7.8% boundary

These are boundary neurons whose presynaptic partners lie outside the 217-column region. They project into the network but have in-degree zero themselves, so no weight vector $\mathbf{w}_i$ needs to be recovered for them.

Among the 13,697 postsynaptic neurons, 10,295 have in-degree ≤ 45 and contribute nothing to the degeneracy. The remaining 3,402 neurons (2,843 with in-degree 46–100 and 559 with in-degree 101–208) account for all **115,223** null dimensions. This means 26.5% of the 434,112 edge weights can be changed freely without affecting the dynamics, while the remaining 318,889 (73.5%) are identifiable.

2.3 Step 3: Within-type degeneracy (the dominant mechanism)

Step 2 showed *how large* the null space is (115,223 dimensions at rank 45) using only linear algebra and SVD. Step 3 explains *why* it is so large: the columnar organization of the connectome forces same-type neurons to have nearly identical activity, which is exactly what makes the columns of $\mathbf{H}_i$ linearly dependent and rank-deficient.

Each of the 217 columns covers a different position in the visual field. For instance, L1 in column 5 sees the light at position (5, 3); L1 in column 6 sees position (6, 3). They receive different stimuli—but when a moving bar sweeps across the eye, column 6 sees what column 5 saw a moment earlier. Over thousands of frames with many stimuli, same-type neurons in different columns produce time-shifted copies of the same signal. Their activity is not identical, but highly correlated.

Now consider a postsynaptic neuron i that receives input from k copies of the same cell type across k neighboring columns. Because these k neurons differ only by a small spatial shift, their activity vectors $\mathbf{h}_{j_1}(t), \dots, \mathbf{h}_{j_k}(t)$ are strongly correlated—like L1 in column 5 vs L1 in column 6, which differ by roughly one timestep. In the matrix $\mathbf{H}_i$, these k columns carry nearly the same information, so their rank is ≈ 1 instead of k , contributing $k - 1$ dimensions to $\ker(\mathbf{H}_i)$.

Why the sum must be constant. In the idealized case where the k columns are identical ($\mathbf{h}_{j_1} = \dots = \mathbf{h}_{j_k} = \mathbf{h}$), apply Step 2’s null space condition $\mathbf{H}_i \boldsymbol{\delta} = \mathbf{0}$. The contribution of these k edges is:

$$\delta_{j_1} \mathbf{h} + \delta_{j_2} \mathbf{h} + \dots + \delta_{j_k} \mathbf{h} = \left(\sum_{m=1}^k \delta_{j_m} \right) \mathbf{h} = \mathbf{0} \quad (9)$$

Since $\mathbf{h} \neq \mathbf{0}$, this requires $\sum_{m=1}^k \delta_{j_m} = 0$. So among k same-type edges, one constraint fixes the sum, leaving $k - 1$ free δ values—these are the null directions counted in Step 4.

2.4 Step 4: Counting the free parameters

For each postsynaptic neuron i and each presynaptic cell type α with $k_{i\alpha} > 1$ incoming edges, the null space gains $k_{i\alpha} - 1$ dimensions. Summing over all neurons and types:

$$\dim \ker(\mathbf{H}) = \sum_{i=1}^N \sum_{\alpha: k_{i\alpha} > 1} (k_{i\alpha} - 1) \quad (10)$$

For the flyvis connectome (13,741 neurons, 434,112 edges, 65 cell types):

Quantity	Value
Total edges E	434,112
Degenerate groups (dst, src_type with $k > 1$)	$\sim 121,000+$
Total null space dimension	$\sim 121,100$
Fraction of all edges in null space	$\sim 28\%$
Identifiable edge parameters	$\sim 313,000$

This means **121,100 weight parameters can be changed freely without any effect on the dynamics**. The solution manifold is not a point but a $\sim 121,100$ -dimensional affine subspace of $\mathbb{R}^E$. Since each null direction admits a continuous scaling $\delta \rightarrow \lambda\delta$ for $\lambda \in \mathbb{R}$, the set of solutions is **uncountably infinite**.

Comparing the two estimates. The SVD-based bound from Step 2 gives **115,223** null dimensions at activity rank 45. The structural within-type count gives $\sim$ **121,100**. The within-type count is slightly larger because it assumes rank ≈ 1 per type group (Step 3’s idealization), while the SVD-based bound uses the global rank 45 which is less conservative for types with fewer than 45 same-type inputs. The two estimates are within $\sim 5\%$ of each other, confirming that within-type degeneracy is the dominant mechanism.

3 Empirical Verification

If the null space analysis (Steps 2–4) is correct, **then** we can change individual weights substantially while preserving the dynamics—as long as the perturbation δ satisfies the sum-zero constraint ($\sum_{m=1}^k \delta_{j_m} = 0$) within each same-type group. Concretely: a perturbed matrix $\mathbf{W} + \delta$ with low connectivity R^2 (weights look very different) should still produce high rollout R^2 (dynamics look the same).

The flyvis connectome has 65 cell types, of which 52 are non-retina types that receive lateral inputs from same-type neurons across columns. Step 3 predicts two regimes depending on how correlated the same-type activity is:

- Types with **identical** same-type activity (within-column projections only) $\rightarrow$ **exact** null space, dynamics unchanged at any perturbation scale
- Types with **correlated but not identical** activity (cross-column projections) $\rightarrow$ **approximate** null space, small divergence growing with perturbation scale

Protocol. For each of the 52 non-retina types, we generated sum-preserving random perturbations at 15 scales (from $\lambda = 0.05$ to 8.0), producing 780 degenerate connectivity matrices. Each was rolled out through the full ODE from identical initial conditions and stimulus.

3.1 Key results

Metric	Value
Variants tested	780
Conn. R^2 range	0.28–1.00
Rollout R^2 range	0.80–1.00
Variants with conn. $R^2 < 0.9$ and rollout $R^2 > 0.99$	30 / 31 (97%)
Variants with conn. $R^2 < 0.99$ and rollout $R^2 > 0.999$	116 / 152 (76%)
Types with exact degeneracy (RMSE $\sim 10^{-7}$)	5 / 52
Types with approximate degeneracy	47 / 52

3.2 All 52 cell types at maximum perturbation ($\lambda = 8.0$)

Color code: **green** $R^2 > 0.9$, **orange** $R^2 > 0.5$, **red** $R^2 \leq 0.5$.

Cell type	Conn. R^2	Rollout R^2	Cell type	Conn. R^2	Rollout R^2
Tm20	0.282	0.997	Mi3	0.733	0.995
Tm5Y	0.566	0.998	Tm1	0.738	1.000
Mi1	0.617	0.998	Mi15	0.741	0.997
Lawf2	0.727	0.995	Tm4	0.753	0.986
Tm16	0.814	1.000	L3	0.871	0.999
Mi4	0.877	0.999	Lawf1	0.899	1.000
R1	0.915	0.975	R6	0.927	1.000
Tm5a	0.933	1.000	R2	0.938	0.998
TmY4	0.941	1.000	Mi13	0.942	1.000
R5	0.951	1.000	TmY5a	0.952	0.999
L2	0.965	1.000	T3	0.966	1.000
T4a	0.967	0.999	Mi14	0.967	1.000
T2a	0.969	1.000	T2	0.969	0.993
Tm2	0.969	1.000	TmY13	0.971	—
TmY14	0.972	1.000	TmY9	0.972	0.960
T4d	0.978	0.999	T4c	0.988	0.989
Tm3	0.990	1.000	T5c	0.990	1.000
T4b	0.990	1.000	T5d	0.992	1.000
Mi2	0.992	1.000	TmY3	0.993	1.000
TmY15	0.993	1.000	TmY10	0.994	1.000
Tm5b	0.995	1.000	Mi11	0.996	0.998
Tm30	0.997	1.000	R3	0.997	1.000
T5a	0.998	1.000	T5b	0.998	1.000
TmY18	0.998	1.000	L5	0.999	1.000*
L4	1.000	1.000*	R7	1.000	1.000*
R8	1.000	1.000*	Tm5c	1.000	1.000*

* Exact degeneracy: RMSE $\sim 10^{-7}$ (machine precision). — = diverged (TmY13 outlier).

Sorted by connectivity R^2 (ascending = most degenerate first). The key pattern: the connectivity column shows **red** and **orange** (weights changed substantially) while the rollout column is entirely **green** (dynamics preserved). The 5 bold types at the bottom have exact degeneracy: their same-type columns in $\mathbf{H}_i$ are exactly identical, so $\mathbf{H}_i\boldsymbol{\delta} = \mathbf{0}$ holds exactly.

3.3 Two regimes

Exact degeneracy (5 types): L4, L5, R7, R8, Tm5c—within-column projections only, no spatial shift. The k same-type presynaptic neurons have truly identical activity.

Approximate degeneracy (47 types): connectivity R^2 ranges from 0.28 (Tm20, 72% of weight variance changed) down to near 1.0, yet rollout R^2 stays above 0.96 in all cases. These results confirm the null space analysis empirically: weights change drastically while dynamics remain nearly identical. The small residual divergence reflects the approximate nature of the null space— $\mathbf{H}_i\boldsymbol{\delta} \approx \mathbf{0}$, not exactly zero—due to the spatial shifts discussed in Step 3.

3.4 Per-scale summary

Each perturbation is $\mathbf{W} + \lambda\boldsymbol{\delta}$, where $\boldsymbol{\delta}$ is a random sum-preserving null direction and λ controls how far we move from the original weights. Larger λ means more different weights, lower connectivity R^2 .

λ	Conn. R^2 range	Mean rollout R^2
0.05	1.000–1.000	1.000000
0.50	0.997–1.000	0.999996
1.00	0.989–1.000	0.999984
2.00	0.955–1.000	0.999936
4.00	0.821–1.000	0.999722
8.00	0.282–1.000	$\sim 0.999^*$

* excluding one outlier (type 57, rollout $R^2 = 0.80$).

4 Discussion

4.1 The inverse problem is fundamentally ill-posed

Recovering the connectivity matrix $\mathbf{W}$ from dynamics alone is ill-posed, even with perfect noise-free observations, known τ_i , V_i^{rest} , known activation function, and known graph topology.

The null space analysis (Steps 2–4) quantifies the ill-posedness: **$\sim 121,100$ out of $434,112$ edge weights** ($\sim 28\%$) can be changed freely without affecting the dynamics. The empirical verification confirms this across all 52 non-retina cell types: at $\lambda = 8.0$, connectivity R^2 drops as low as 0.28 while rollout R^2 stays above 0.96.

The set of valid solutions is **uncountably infinite**. For each of the 52 cell types independently, the perturbation δ can be scaled by any $\lambda \in \mathbb{R}$, and different types can use different λ values. The solution manifold is not a point but a $\sim 121,100$ -dimensional continuous subspace of $\mathbb{R}^{434,112}$.

The ill-posedness is **structural**, arising from the columnar organization of the circuit: same-type neurons in different columns produce strongly correlated activity (Step 3), making their individual contributions to a shared target indistinguishable from dynamics alone.

4.2 Implications for GNN-based inference

The GNN approximates the dynamics as:

$$\frac{dv_i(t)}{dt} = f_\theta\left(v_i(t), \mathbf{a}_i, \sum_{j \in \mathcal{N}_i} \widehat{\mathbf{W}}_{ij} g_\phi(v_j(t), \mathbf{a}_j)^2, I_i(t)\right) \quad (11)$$

where $f_\theta = \text{MLP}_0$ and $g_\phi = \text{MLP}_1$ are three-layer perceptrons, $\widehat{\mathbf{W}}_{ij}$ are learnable edge weights, and $\mathbf{a}_i \in \mathbb{R}^2$ are learnable per-neuron embeddings. The training minimizes:

$$\mathcal{L}_{\text{pred}} = \sum_{i,t} \|\widehat{y}_i(t) - y_i(t)\|_2 \quad (12)$$

between simulator $y_i(t) = dv_i(t)/dt$ and GNN predictions $\widehat{y}_i(t) = \widehat{dv}_i(t)/dt$, augmented with regularization:

$$\begin{aligned} \mathcal{L} = & \|\widehat{\mathbf{y}} - \mathbf{y}\|_2 + \lambda_0 \|\theta\|_1 + \lambda_1 \|\phi\|_1 + \lambda_2 \|\widehat{\mathbf{W}}\|_1 \\ & + \gamma_0 \|\theta\|_2 + \gamma_1 \|\phi\|_2 \\ & + \mu_0 \left\| \text{ReLU}\left(-\frac{\partial g_\phi(v, \mathbf{a})}{\partial v}\right) \right\|_2 + \mu_1 \|g_\phi(v_\star, \mathbf{a}) - v_\star\|_2 \end{aligned} \quad (13)$$

where λ terms promote sparsity, γ terms stabilize learned functions, μ_0 enforces monotonicity of the edge message, and μ_1 normalizes with respect to reference voltage $v_\star$.

When the GNN achieves $R_{\mathbf{W}}^2 = 0.997$ (as our LLM-optimized model does at $\sigma = 0.5$), it is recovering the $\sim 313,000$ identifiable parameters well while the $\sim 121,100$ null-space parameters are essentially free. The GNN’s implicit inductive bias (message-passing, weight sharing) acts as an implicit regularizer that selects a particular solution from the uncountable solution manifold.

4.3 Why noise helps

Adding process noise $\sigma \eta_i(t)$ to the simulation:

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij} \text{ReLU}(v_j(t)) + I_i(t) + \sigma \eta_i(t) \quad (14)$$

breaks the exact within-type degeneracy because:

1. Each neuron receives independent noise realizations, even if same-type
2. This injects independent variation into the columns of $\mathbf{H}_i$, breaking the within-type correlations from Step 3
3. The effective rank of $\mathbf{H}_i$ increases from 45 (noise-free) toward d_i
4. The null space $\dim \ker(\mathbf{H}_i) = d_i - \text{rank}(\mathbf{H}_i)$ shrinks, making more edge weights identifiable

Quantitative measurement. We measure how noise changes the effective rank of the population activity, and thus the null space dimension, using the same formula as Step 2:

$$\text{total null} = \sum_{i=1}^{N_{\text{post}}} \max(0, d_i - r) \quad (15)$$

SVD methodology. For each noise level ($\sigma = 0, 0.05, 0.5$), we:

1. Load the full voltage trace $\mathbf{V} \in \mathbb{R}^{64000 \times 13741}$ and subsample to 8,000 timesteps (every 8th frame)
2. Apply $h_j(t) = \text{ReLU}(v_j(t))$ to get the activity matrix $\mathbf{H} \in \mathbb{R}^{8000 \times 13741}$
3. Subsample to every 14th neuron $\rightarrow \mathbf{H}_{\text{sub}} \in \mathbb{R}^{8000 \times 982}$
4. Compute the **full** SVD of $\mathbf{H}_{\text{sub}}$ (all 982 singular values), so the total variance $\sum_k s_k^2$ is exact
5. Find the smallest r such that $\sum_{k=1}^r s_k^2 \geq 0.99 \cdot \sum_k s_k^2$
6. Plug r into the per-neuron null space formula above, using the in-degree distribution from the flyvis connectome (434,112 edges, max in-degree 208)

Results. The $\sigma = 0$ row reproduces Step 2’s estimate (rank 45, null 115,223) using the same SVD methodology. The noisy conditions use identical subsampling, so the ranks are directly comparable. GNN experimental results are from `flyvis_results.md` (10 seeds each):

Noise	Activity rank r	Null space dim	% identifiable	Observed $R_{\mathbf{W}}^2$
$\sigma = 0$	45	115,223	73.5%	0.899 ± 0.030
$\sigma = 0.05$	134	10,227	97.6%	0.962 ± 0.012
$\sigma = 0.5$	781	0	100.0%	0.997 ± 0.000

The trend validates the mechanism: noise increases the effective rank monotonically ($45 \rightarrow 134 \rightarrow 781$), shrinking the null space and making more weights identifiable. At $\sigma = 0.5$, the rank (781) exceeds the maximum in-degree (208), so every neuron’s incoming weights are fully identifiable—the null space vanishes entirely, consistent with $R_{\mathbf{W}}^2 = 0.997$.

Alternatively, to break the strong within-type correlations experimentally, one could use optogenetics to inject independent perturbations into each neuron of a given type (e.g., stimulate each L1 with a different random light pattern), or inversely silence a subset of same-type neurons across columns (e.g., ablate L1 in columns 4 and 6 but not 5). Both approaches make the columns of $\mathbf{H}_i$ distinguishable, increasing $\text{rank}(\mathbf{H}_i)$ and shrinking the null space—the same mechanism as process noise, but achievable in real tissue.

5 Summary

Recovering the connectome from neural dynamics is fundamentally ill-posed: the columnar organization of the flyvis circuit forces same-type neurons across columns to produce strongly correlated activity, making their individual synaptic contributions indistinguishable. This structural degeneracy creates a $\sim 115,000$ -dimensional null space (26.5% of all 434,112 edge weights), confirmed empirically by generating 780 perturbed connectivity matrices where weights change drastically (conn. R^2 as low as 0.28) while dynamics remain nearly identical (rollout $R^2 > 0.96$). Adding independent process noise breaks the within-type correlations and increases the effective

activity rank from 45 (noise-free) to 781 ($\sigma = 0.5$), eliminating the null space entirely and enabling the GNN to achieve $R_{\mathbf{W}}^2 = 0.997$. The analysis provides both a theoretical framework (SVD-based null space bounds) and a mechanistic explanation for why biological noise—intrinsic to neural circuits—naturally improves connectome identifiability.